Statistics & Probability

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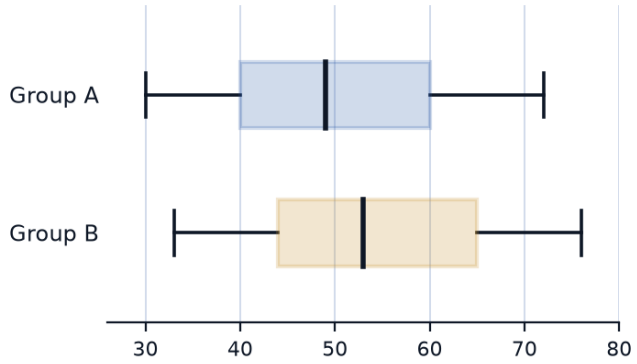
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Mean, median, and mode

Find the mean of $4, 8, 6, 10, 7$.

2 [Foundational] • Target: Mean, median, and mode

Find the median of $3, 9, 5, 1, 7$.



3 [Foundational] • Target: Mean, median, and mode

Find the mode of $2, 4, 4, 6, 8, 4, 2$.

4 [Foundational] • Target: Mean, median, and mode

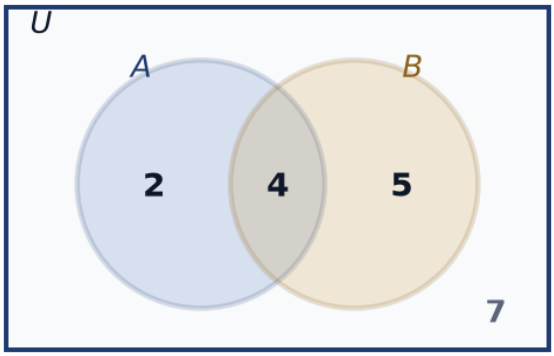
A data set has values $8, 12, 10, 15, 10, 14, 11$. Find the mean, median, and mode.

5 [Foundational] • Target: Range and quartiles

Find the range of $5, 12, 8, 20, 3$.

6 [Foundational] • Target: Range and quartiles

For the data set $2, 4, 5, 7, 8, 9, 12$, find Q_1 and Q_3.



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7 [Foundational] • Target: Range and quartiles

Find the interquartile range (IQR) for the same data set.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Variance and standard deviation

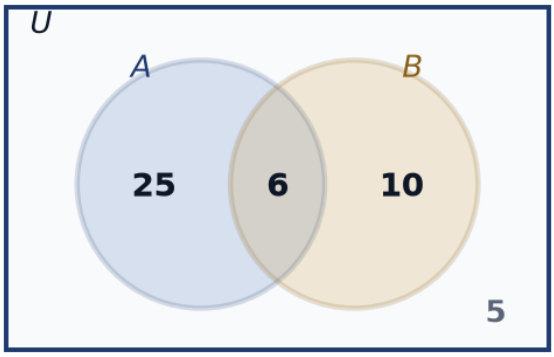
Find the variance of $2, 4, 6, 8, 10$, using $\sigma^2 = \frac{\sum(x-\mu)^2}{n}$.

9 [Intermediate] • Target: Variance and standard deviation

Find the standard deviation of the same data set, rounded to two decimal places.

10 [Intermediate] • Target: Variance and standard deviation

A data set has variance 25. State its standard deviation.



11 [Intermediate] • Target: Frequency tables

A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

12 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the mode.

13 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the median.

14 [Intermediate] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

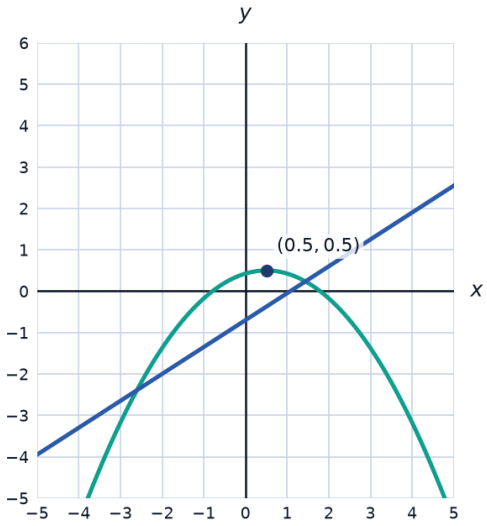
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15 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Exponentials & Logarithms

A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

16 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Algebra & Equations

A card is drawn from a standard 52-card deck. Find $P(\text{king})$.



17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Functions & Transformations

Two fair coins are flipped. Find $P(\text{both heads})$.

18 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 3.01 Sequences & Series

A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?

19 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 4.01 Trigonometry

If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

20 [Challenging] • Target: Independent events • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Two independent events have $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Frequency tables

A frequency table shows value 1 with frequency 4, value 2 with frequency 6, and value 3 with frequency 5. Find the total number of data points.

a. 10

b. 15

c. 3

d. 6

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22 [Foundational] • Target: MCQ — Frequency tables

Using the same frequency table, find the mean.

a. 2.33

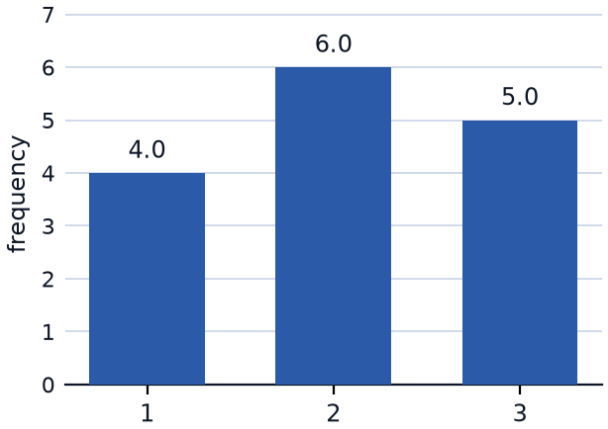
b. ≈ 2.07

c. 2.00

d. 1.87

23 [Foundational] • Target: MCQ — Frequency tables

The bar chart shows a frequency table. Find the total number of data points.



a. 15

b. 6

c. 3

d. 10

24 [Foundational] • Target: MCQ — Conditional probability

A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$

c. $\frac{5}{12}$

d. $\frac{4}{12}$

25 [Foundational] • Target: MCQ — Counting: permutations and combinations

In how many ways can 6 people be arranged in a line?

a. 720

b. 120

c. 36

d. 6

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26 [Foundational] • Target: MCQ — The five-number summary

For the data set $10, 15, 20, 25, 30, 35, 40$, find the median.

a. 25

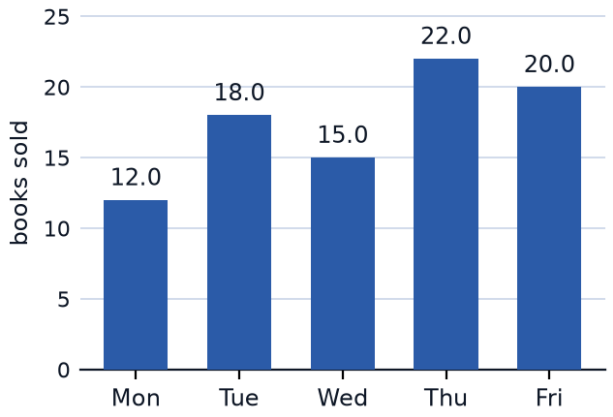
b. 30

c. 20

d. 27.5

27 [Foundational] • Target: MCQ — Reading data from a bar chart

The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.



a. 87

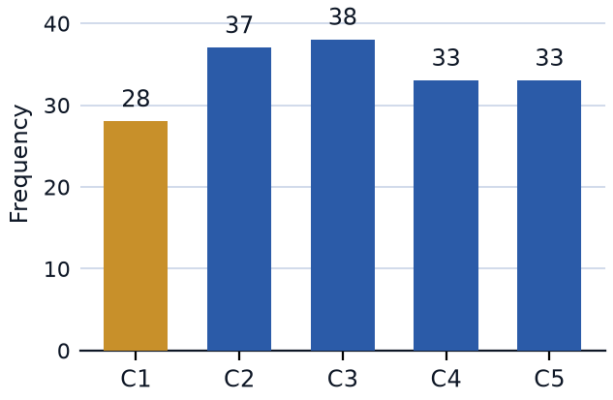
b. 22

c. 77

d. 97

28 [Foundational] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, find the mean number of books sold per day.



a. 15

b. 18

c. 20

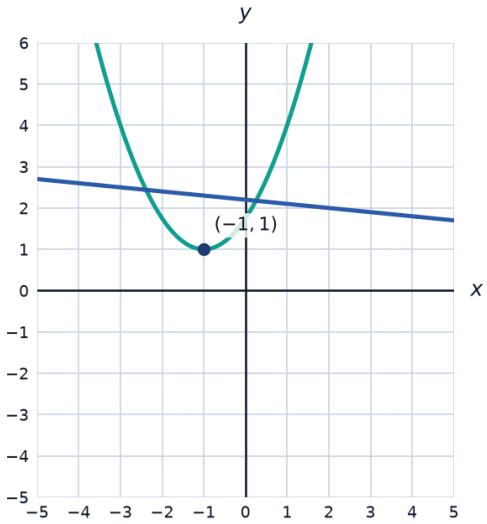
d. 17.4

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: MCQ — Reading data from a bar chart**

Using the same bar chart, find the range of books sold.



a. 12

b. 22

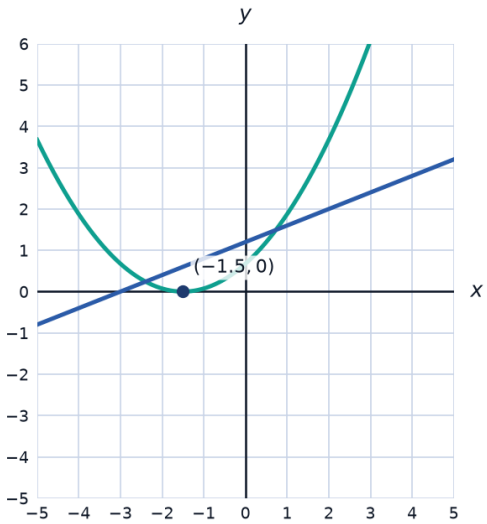
c. 8

d. 10

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30 [Intermediate] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, on which day were the most books sold?



a. Wednesday

b. Thursday

c. Tuesday

d. Friday

31 [Intermediate] • Target: MCQ — Word problems: statistics and probability in context

A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

a. 0.84

b. 0.6

c. 0.16

d. 1.0

32 [Intermediate] • Target: MCQ — Word problems: statistics and probability in context

A quality-control inspector finds that 3% of items from a production line are defective. If 500 items are inspected, how many are expected to be defective?

a. 150

b. 15

c. 3

d. 30

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33 [Intermediate] • Target: MCQ — Word problems: statistics and probability in context

In a survey of 180 people, 108 prefer coffee over tea. Find the probability, as a percent, that a randomly selected person prefers coffee.

a. 60%

b. 50%

c. 108%

d. 40%

34 [Intermediate] • Target: MCQ — Range and quartiles

A code uses 3 distinct digits from 0 to 9, where order matters. How many possible codes are there?

a. 710

b. 730

c. 1000

d. 720

35 [Intermediate] • Target: MCQ — Variance and standard deviation

[illegible] (G4). In a study of the number of hours high school students spend studying per week, the following data was collected and represented in a bar chart. If the mean number of study hours for all students is 20 hours, find the value of x which represents the study hours for Group 4 (G4).

a. 20

b. 22

c. 24

d. 26

36 [Intermediate] • Target: MCQ — Frequency tables

[illegible] The bar chart above shows the distribution of student scores on a test. Which score range did the largest number of students achieve?

a. 60-69

b. 70-79

c. 80-89

d. 90-99

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37 [Intermediate] • Target: MCQ — Basic probability

[redacted] 100 [redacted] [redacted] [redacted]

39 [Challenging] • Target: MCQ — Independent events • Cross-Lesson: 4.01 Trigonometry

[illegible] A ■ C■ The bar chart below shows the number of students who won gold medals in different competitions. What is the total number of gold medals won by students in competitions A and C?

a. 30

b. 34

c. 40

d. 44

40 [Challenging] • Target: MCQ — Conditional probability • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

[illegible] The following bar chart shows the number of students who prefer different fruits. Which fruit is the most preferred?

a. ■■■■■■

b. ■■■■■

c. ■■■■■■■■

d. ■■■■■■■■

41 [Challenging] • Target: MCQ — Counting: permutations and combinations • Cross-Lesson: 6.01 Limits & Derivatives

■■ ■■■■■■■■ ■■■■■■■■■ ■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■■ ■■■■■ ■■■■ ■■■■ $\mu + \sigma$ ■ $\mu + 2\sigma$■ Percentage of data between 1 and 2 standard deviations above mean?

a. 13.5%

b. 34%

c. 2%

d. 68%

42 [Challenging] • Target: MCQ — The five-number summary • Cross-Lesson: 7.01 Integration & Applications

■■ ■■■■■■■■■ ■■■■■■■■■ ■■■■■■: $2, 4, 6, 8, 10$ ■■■■■■■ Sample standard deviation of {2, 4, 6, 8, 10}?

a. $\sqrt{10}$

b. $\sqrt{8}$

c. 3

d. $\sqrt{5}$

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43 [Challenging] • Target: MCQ — Reading data from a bar chart • Cross-Lesson: 8.01 Exponentials & Logarithms

[redacted] $P(A) = 0.4$ [redacted] $P(B) = 0.5$ [redacted] $P(A \cup B)$[redacted]

Find P(A or B) for independent events.

a. 0.7

b. 0.9

c. 0.2

d. 0.6

44 [Challenging] • Target: MCQ — Word problems: statistics and probability in context • Cross-Lesson: 1.01 Algebra & Equations

[redacted]: 2, 5, 8, 10, 14, 18[redacted] Find the median of an even number of values.

a. 9

b. 8

c. 10

d. 8.5

45 [Challenging] • Target: Mean, median, and mode • Cross-Lesson: 2.01 Functions & Transformations

[redacted] A [redacted] B [redacted] $P(A \cap B)$ [redacted]: If A and B are mutually exclusive, P(A and B) = ?

a. 0

b. 1

c. $P(A) \cdot P(B)$

d. $P(A) + P(B)$

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